DBMS Experiential Learning – Level 3 Project

Team Members

Name: Aaditya Vashisht
SRN: PES1UG24CS004

Name: Ananya Uppili
SRN: PES1UG24CS060

Project title : Resort Management System

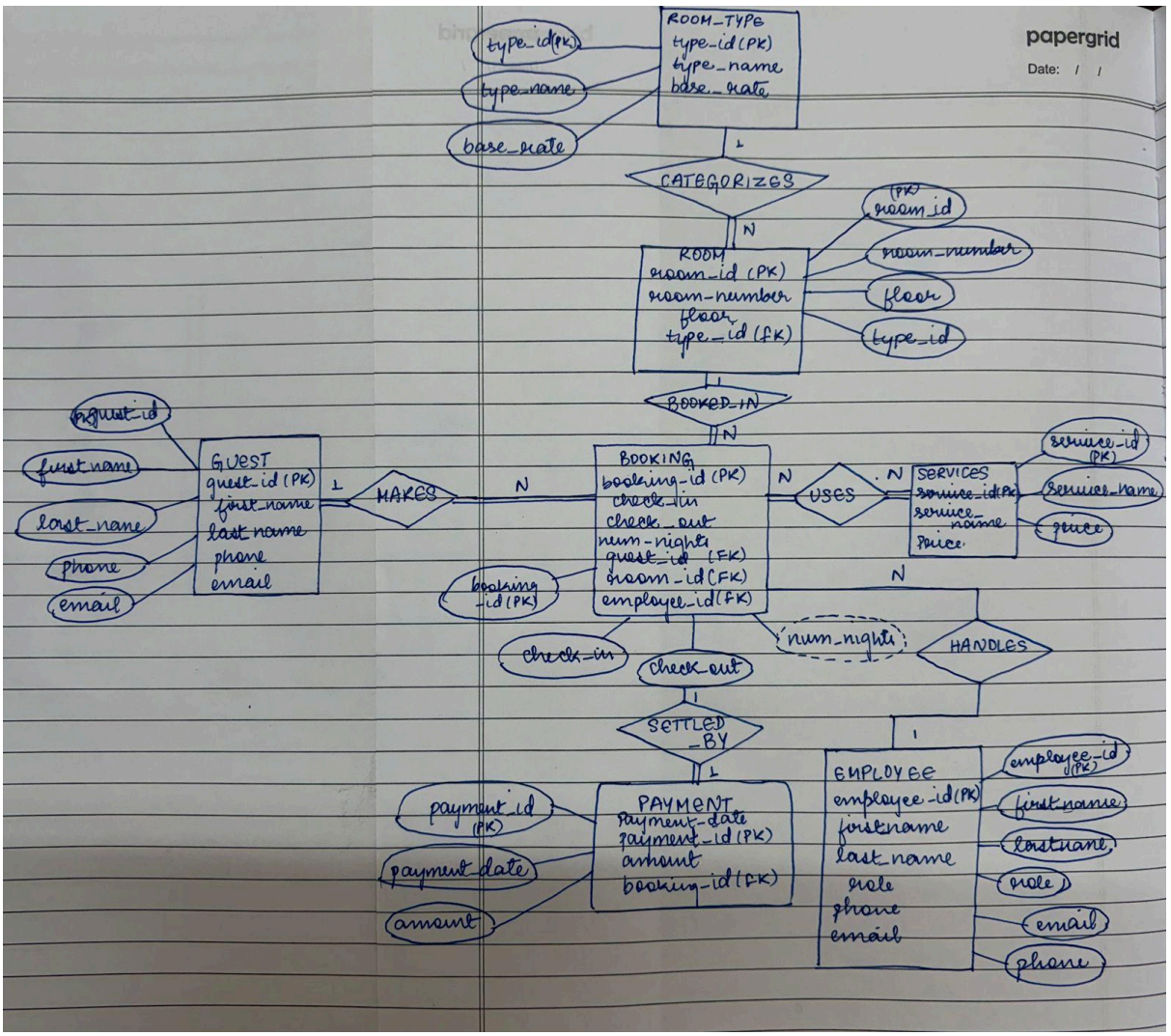
Problem Statement

The Resort Management System is a database system that helps a resort manage its day-to-day operations in one place. It is mainly used by receptionists, resort staff, and managers to handle guest information, room bookings, payments, and additional services. Instead of maintaining these details separately, the system keeps them organized and connected.

The main entities in the system are Guests, Rooms, Room Types, Bookings, Payments, Employees, and Services. A guest can make multiple bookings, and each booking contains details such as the room booked, check-in and check-out dates, and payment status. Guests can also use additional services such as a spa, restaurant, or recreational activities.

The database should be able to answer important queries such as: which rooms are available for specific dates, what bookings a particular guest has made, how much a guest needs to pay including additional services, and how many rooms are occupied or how much revenue the resort generated during a given period.

ER Diagram



Relational Schema

ROOM_TYPE(
  type_id PK,
  type_name,
  base_rate
)

ROOM(
  room_id PK,
  room_number,
  floor,
  type_id FK
)

GUEST(
  guest_id PK,
  first_name,
  last_name,

```
    phone,
    email
)

EMPLOYEE(
    employee_id PK,
    first_name,
    last_name,
    role,
    phone,
    email
)

BOOKING(
    booking_id PK,
    check_in,
    check_out,
    num_nights,
    guest_id FK,
    room_id FK,
    employee_id FK
)

SERVICE(
    service_id PK,
    service_name,
    price
)

PAYMENT(
    payment_id PK,
    payment_date,
    amount,
    booking_id FK
)

BOOKING_SERVICE(
    booking_id PK, FK,
    service_id PK, FK
)
```

```
mysql> CREATE DATABASE resort_entries;
Query OK, 1 row affected (0.01 sec)

mysql> USE resort_entries;
Database changed
mysql>
```

```
mysql> DROP TABLE IF EXISTS ROOM_TYPE;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS GUEST;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS EMPLOYEE;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS SERVICE;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS ROOM;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS BOOKING;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS PAYMENT;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DROP TABLE IF EXISTS BOOKING_SERVICE;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
```

```
mysql> CREATE TABLE ROOM_TYPE (
    -> type_id     INT AUTO_INCREMENT PRIMARY KEY,
    -> type_name   VARCHAR(50)  NOT NULL,
    -> base_rate   DECIMAL(10,2) NOT NULL CHECK (base_rate >= 0)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE ROOM (
    -> room_id      INT AUTO_INCREMENT PRIMARY KEY,
    -> room_number  VARCHAR(10) NOT NULL UNIQUE,
    -> floor        INT NOT NULL,
    -> type_id      INT NOT NULL,
    -> CONSTRAINT fk_room_type
    -> FOREIGN KEY (type_id) REFERENCES ROOM_TYPE(type_id)
    -> ON UPDATE CASCADE
    -> ON DELETE RESTRICT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE GUEST (
    -> guest_id     INT AUTO_INCREMENT PRIMARY KEY,
    -> first_name   VARCHAR(50) NOT NULL,
    -> last_name    VARCHAR(50) NOT NULL,
    -> phone        VARCHAR(20),
    -> email        VARCHAR(100) UNIQUE
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE EMPLOYEE (
    -> employee_id  INT AUTO_INCREMENT PRIMARY KEY,
    -> first_name   VARCHAR(50) NOT NULL,
    -> last_name    VARCHAR(50) NOT NULL,
    -> role         VARCHAR(50) NOT NULL,
    -> phone        VARCHAR(20),
    -> email        VARCHAR(100) UNIQUE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE SERVICE (
    -> service_id    INT AUTO_INCREMENT PRIMARY KEY,
    -> service_name  VARCHAR(100) NOT NULL,
    -> price         DECIMAL(10,2) NOT NULL CHECK (price >= 0)
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE BOOKING (
    ->     booking_id   INT AUTO_INCREMENT PRIMARY KEY,
    ->     check_in     DATE NOT NULL,
    ->     check_out    DATE NOT NULL,
    ->     guest_id     INT NOT NULL,
    ->     room_id      INT NOT NULL,
    ->     employee_id  INT NOT NULL,
    ->     CONSTRAINT chk_booking_dates CHECK (check_out > check_in),
    ->     CONSTRAINT fk_booking_guest
    ->         FOREIGN KEY (guest_id) REFERENCES GUEST(guest_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_booking_room
    ->         FOREIGN KEY (room_id) REFERENCES ROOM(room_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_booking_employee
    ->         FOREIGN KEY (employee_id) REFERENCES EMPLOYEE(employee_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT
    -> );
Query OK, 0 rows affected (1.04 sec)
```

```
mysql> CREATE TABLE BOOKING_SERVICE (
    ->     booking_id   INT NOT NULL,
    ->     service_id   INT NOT NULL,
    ->     PRIMARY KEY (booking_id, service_id),
    ->     CONSTRAINT fk_bs_booking
    ->         FOREIGN KEY (booking_id) REFERENCES BOOKING(booking_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->     CONSTRAINT fk_bs_service
    ->         FOREIGN KEY (service_id) REFERENCES SERVICE(service_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE RESTRICT
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE PAYMENT (
    ->     payment_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     payment_date  DATE NOT NULL,
    ->     amount        DECIMAL(10,2) NOT NULL CHECK (amount >= 0),
    ->     booking_id    INT NOT NULL UNIQUE,
    ->     CONSTRAINT fk_payment_booking
    ->         FOREIGN KEY (booking_id) REFERENCES BOOKING(booking_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE INDEX idx_room_type_id       ON ROOM(type_id);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_booking_guest_id   ON BOOKING(guest_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_booking_room_id    ON BOOKING(room_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_booking_emp_id     ON BOOKING(employee_id);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC EMPLOYEE;
+-------------+-------------+------+-----+---------+----------------+
| Field       | Type        | Null | Key | Default | Extra          |
+-------------+-------------+------+-----+---------+----------------+
| employee_id | int(11)     | NO   | PRI | NULL    | auto_increment |
| first_name  | varchar(50) | NO   |     | NULL    |                |
| last_name   | varchar(50) | NO   |     | NULL    |                |
| role        | varchar(50) | NO   |     | NULL    |                |
| phone       | varchar(20) | YES  |     | NULL    |                |
| email       | varchar(100)| YES  | UNI | NULL    |                |
+-------------+-------------+------+-----+---------+----------------+
6 rows in set (0.03 sec)

mysql> DESC ROOM_TYPE;
+-----------+---------------+------+-----+---------+----------------+
| Field     | Type          | Null | Key | Default | Extra          |
+-----------+---------------+------+-----+---------+----------------+
| type_id   | int(11)       | NO   | PRI | NULL    | auto_increment |
| type_name | varchar(50)   | NO   |     | NULL    |                |
| base_rate | decimal(10,2) | NO   |     | NULL    |                |
+-----------+---------------+------+-----+---------+----------------+
3 rows in set (0.01 sec)

mysql> DESC ROOM;
+-------------+-------------+------+-----+---------+----------------+
| Field       | Type        | Null | Key | Default | Extra          |
+-------------+-------------+------+-----+---------+----------------+
| room_id     | int(11)     | NO   | PRI | NULL    | auto_increment |
| room_number | varchar(10) | NO   | UNI | NULL    |                |
| floor       | int(11)     | NO   |     | NULL    |                |
| type_id     | int(11)     | NO   | MUL | NULL    |                |
+-------------+-------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)
```

```
mysql> DESC GUEST;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| guest_id   | int(11)      | NO   | PRI | NULL    | auto_increment |
| first_name | varchar(50)  | NO   |     | NULL    |                |
| last_name  | varchar(50)  | NO   |     | NULL    |                |
| phone      | varchar(20)  | YES  |     | NULL    |                |
| email      | varchar(100) | YES  | UNI | NULL    |                |
+------------+--------------+------+-----+---------+----------------+
5 rows in set (0.01 sec)

mysql> DESC BOOKING;
+-------------+---------+------+-----+---------+----------------+
| Field       | Type    | Null | Key | Default | Extra          |
+-------------+---------+------+-----+---------+----------------+
| booking_id  | int(11) | NO   | PRI | NULL    | auto_increment |
| check_in    | date    | NO   |     | NULL    |                |
| check_out   | date    | NO   |     | NULL    |                |
| guest_id    | int(11) | NO   | MUL | NULL    |                |
| room_id     | int(11) | NO   | MUL | NULL    |                |
| employee_id | int(11) | NO   | MUL | NULL    |                |
+-------------+---------+------+-----+---------+----------------+
6 rows in set (0.01 sec)

mysql> DESC SERVICE;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| service_id   | int(11)       | NO   | PRI | NULL    | auto_increment |
| service_name | varchar(100)  | NO   |     | NULL    |                |
| price        | decimal(10,2) | NO   |     | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
3 rows in set (0.08 sec)
```

```
mysql> DESC SERVICE;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| service_id   | int(11)       | NO   | PRI | NULL    | auto_increment |
| service_name | varchar(100)  | NO   |     | NULL    |                |
| price        | decimal(10,2) | NO   |     | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
3 rows in set (0.08 sec)

mysql> DESC PAYMENT;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| payment_id   | int(11)       | NO   | PRI | NULL    | auto_increment |
| payment_date | date          | NO   |     | NULL    |                |
| amount       | decimal(10,2) | NO   |     | NULL    |                |
| booking_id   | int(11)       | NO   | UNI | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
4 rows in set (0.06 sec)

mysql> DESC BOOKING_SERVICE;
+------------+---------+------+-----+---------+-------+
| Field      | Type    | Null | Key | Default | Extra |
+------------+---------+------+-----+---------+-------+
| booking_id | int(11) | NO   | PRI | NULL    |       |
| service_id | int(11) | NO   | PRI | NULL    |       |
+------------+---------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```